



Web Programming
Lecture 5

Bootstrap + Tailwind

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What is bootstrap?

- Bootstrap is a powerful, open-source front-end framework used for designing responsive and mobile-first websites.
- Why use bootstrap?
 - Speeds up development.
 - Pre-styled components.
 - Easy to create responsive designs.

Key Features of Bootstrap



- **Responsive Grid System**
- **Pre-designed Components**
 - Buttons, Modals, Alerts, etc.
- **Customizable**
 - Built with Sass and a theming system.
- **Cross-Browser Compatibility**
- **Extensive Documentation**



Bootstrap 5

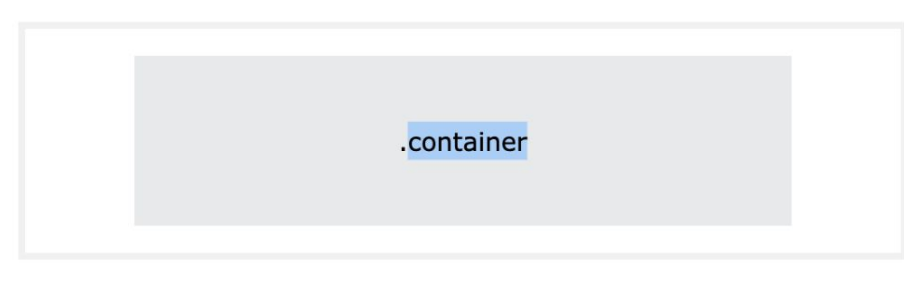
```
<!-- Bootstrap CSS --> <link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/  
bootstrap.min.css" rel="stylesheet"  
integrity="sha384-EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTWFs  
pd3yD65VohhpuuCOMLASjC" crossorigin="anonymous">
```

```
<!-- Bootstrap JS Bundle with Popper --> <script  
src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bo  
otstrap.bundle.min.js"  
integrity="sha384-MrcW6ZMFYIzcla8NI+NtUVF0sA7MsXsP1UyJoMp4YL  
EuNSfAP+JcXn/tWtlaxVXM" crossorigin="anonymous"></script>
```

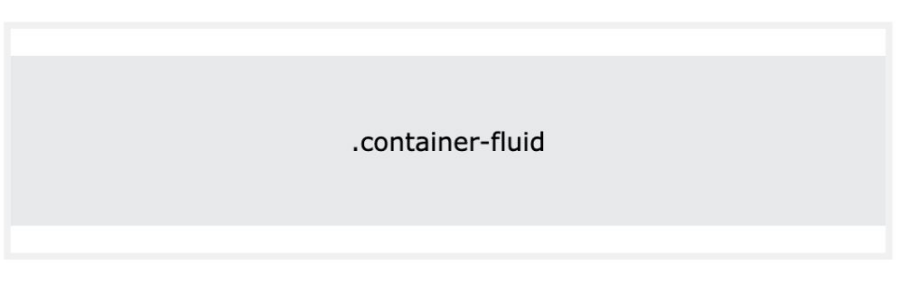
Bootstrap Container

Containers are used to pad the content inside of them, and there are two container classes available:

1. The `.container` class provides a responsive fixed width container
2. The `.container-fluid` class provides a full width container, spanning the entire width of the viewport



`.container`

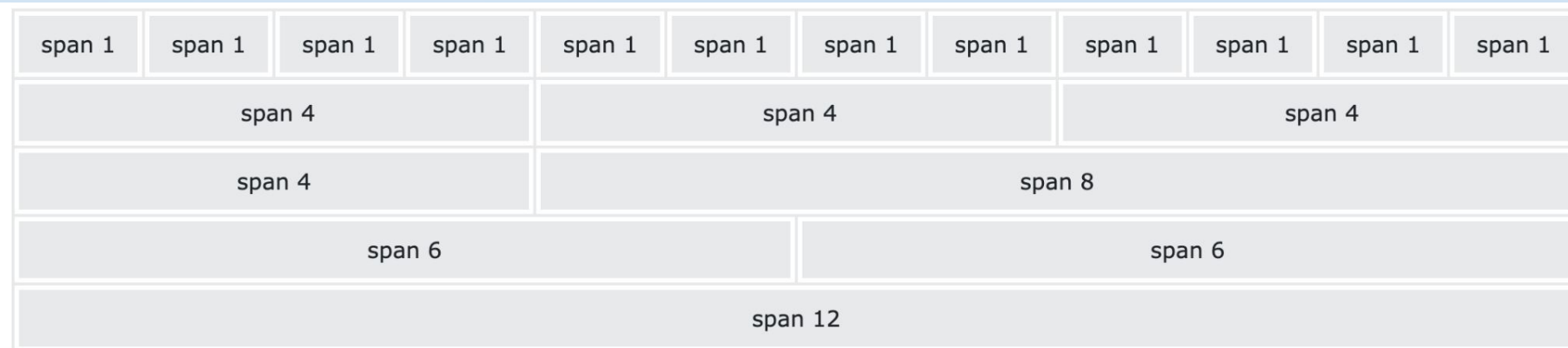


`.container-fluid`



Bootstrap Grid System

- **What is the Grid System?**
 - Divides the page into a 12-column layout.
- **Core Concepts:**
 - Containers (fixed and fluid).
 - Rows and Columns.
 - Breakpoints for responsiveness.





```
<div class="container">  
  <div class="row">  
    <div class="col-md-6"></div>  
    <div class="col-md-6"></div>  
  </div>  
  <div class="row">  
    <div class="col-md-4"></div>  
    <div class="col-md-4"></div>  
    <div class="col-md-4"></div>  
  </div>  
</div>
```

Typography and Components



- **Typography**
 - Predefined styles for headings, paragraphs, and text utilities.
- **Common Components:**
 - Buttons
 - Alerts
 - Navigation Bars
 - Modals
- **Example:**

```
<button class="btn btn-primary">Click Me!</button>
```




Bootstrap Buttons

Primary

Secondary

Success

Danger

Warning

Info

Light

Dark

Link

```
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-dark">Dark</button>

<button type="button" class="btn btn-link">Link</button>
```

Bootstrap Alerts



```
<div class="alert alert-primary" role="alert">
  This is a primary alert—check it out!
</div>
<div class="alert alert-secondary" role="alert">
  This is a secondary alert—check it out!
</div>
<div class="alert alert-success" role="alert">
  This is a success alert—check it out!
</div>
<div class="alert alert-danger" role="alert">
  This is a danger alert—check it out!
</div>
<div class="alert alert-warning" role="alert">
  This is a warning alert—check it out!
</div>
<div class="alert alert-info" role="alert">
  This is a info alert—check it out!
</div>
<div class="alert alert-light" role="alert">
  This is a light alert—check it out!
</div>
<div class="alert alert-dark" role="alert">
  This is a dark alert—check it out!
</div>
```

This is a primary alert—check it out!

This is a secondary alert—check it out!

This is a success alert—check it out!

This is a danger alert—check it out!

This is a warning alert—check it out!

This is a info alert—check it out!

This is a light alert—check it out!

This is a dark alert—check it out!



Bootstrap Navigation Bar

Active Link Link Disabled

```
<nav class="nav">
  <a class="nav-link active" href="#">Active</a>
  <a class="nav-link" href="#">Link</a>
  <a class="nav-link" href="#">Link</a>
  <a class="nav-link disabled" href="#">Disabled</a>
</nav>
```

- **.nav** uses **display: flex**
- **flex-column** class on **nav** makes it vertical

Active

Link

Link

Disabled

```
<ul class="nav flex-column">
  <li class="nav-item">
    <a class="nav-link active" href="#">Active</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#">Disabled</a>
  </li>
</ul>
```

Bootstrap Tabs



Active

Link

Link

Disabled

```
<ul class="nav nav-tabs">
  <li class="nav-item">
    <a class="nav-link active" href="#">Active</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#">Disabled</a>
  </li>
</ul>
```



Bootstrap Dropdown Menu

Copy

```
<div class="dropdown show">
  <a class="btn btn-secondary dropdown-toggle" href="#" role="button" id="dropdownMenuLink" data-toggle="dropdown" aria-
    Dropdown link
  </a>

  <div class="dropdown-menu" aria-labelledby="dropdownMenuLink">
    <a class="dropdown-item" href="#">Action</a>
    <a class="dropdown-item" href="#">Another action</a>
    <a class="dropdown-item" href="#">Something else here</a>
  </div>
</div>
```

Dropdown link ▼

Action

Another action

Something else here

Bootstrap Form



Email address

We'll never share your email with anyone else.

Password

☐ Check me out

Copy

```
<form>
  <div class="form-group">
    <label for="exampleInputEmail1">Email address</label>
    <input type="email" class="form-control" id="exampleInputEmail1" aria-describedby="emailHelp" placeholder="Enter ema
    <small id="emailHelp" class="form-text text-muted">We'll never share your email with anyone else.</small>
  </div>
  <div class="form-group">
    <label for="exampleInputPassword1">Password</label>
    <input type="password" class="form-control" id="exampleInputPassword1" placeholder="Password">
  </div>
  <div class="form-check">
    <input type="checkbox" class="form-check-input" id="exampleCheck1">
    <label class="form-check-label" for="exampleCheck1">Check me out</label>
  </div>
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Bootstrap Responsive

- **Breakpoints:**
 - `sm, md, lg, xl, xxl`
 - Adapt layout based on screen size.
- **Mobile-First Approach:**
 - Prioritize small screens and scale up.
- **Example:**

```
<div class="col-sm-12 col-md-6 col-lg-4">Column 1</div>
```

```
<div class="col-sm-12 col-md-6 col-lg-4">Column 2</div>
```

```
<div class="col-sm-12 col-md-6 col-lg-4">Column 3</div>
```

```
<div class="col-sm-12 col-md-6 col-lg-4">Column 4</div>
```

Try it yourself



Question:

Write complete HTML code using Bootstrap to create a responsive grid layout with 12 boxes that behaves as follows:

Requirements

On mobile phones (small screens) → show 1 column per row

On small screens (phones in landscape / medium devices) → show 2 columns per row

On laptops → show 6 columns per row

Introduction to Tailwind CSS



Tailwind CSS

- A utility-first CSS framework
- Lets us style HTML using predefined utility classes
- No need to write most custom CSS
- Responsive design is built in

```
<button class="bg-blue-500 text-white px-4 py-2 rounded">  
  Click Me  
</button>
```

What Does "Utility-First" Mean?



Each class generally performs **one small styling job**.

```
<div class="text-center mt-4 bg-gray-100 p-6">
```

Class	Meaning
text-center	Center text
mt-4	Margin-top
bg-gray-100	Background color
p-6	Padding

Colors



Tailwind provides predefined color utilities.

`property-color-shade`

`text-red-500 bg-blue-600`
`border-gray-300`

In `text-blue-500`, **500** is the shade/intensity of the blue.

50	→ very light
100	→ light
200	
300	
400	
500	→ medium/default
600	
700	
800	
900	→ very dark
950	→ darkest

Spacing

p-4 → padding
px-4 → padding left + right
py-4 → padding top + bottom

m-4 → margin
mx-4 → margin left + right
my-4 → margin top + bottom

mt-4 → margin-top
mb-4 → margin-bottom

1	→ 0.25rem	→ 4px
2	→ 0.5rem	→ 8px
3	→ 0.75rem	→ 12px
4	→ 1rem	→ 16px
5	→ 1.25rem	→ 20px
6	→ 1.5rem	→ 24px
8	→ 2rem	→ 32px

1rem = 16px



Width, Height & Sizing

w-1/2 → 50%
w-1/3 → 33.333%
w-2/3 → 66.667%

w-1/4 → 25%
w-2/4 → 50%
w-3/4 → 75%

w-1/5 → 20%
w-2/5 → 40%
w-3/5 → 60%
w-4/5 → 80%

w-1/6 → 16.667%
w-2/6 → 33.333%
w-3/6 → 50%
w-4/6 → 66.667%
w-5/6 → 83.333%

w-full → 100% of its parent

w-4 → 1rem → 16px
w-8 → 2rem → 32px
w-16 → 4rem → 64px
w-64 → 16rem → 256px

w-screen =
viewport width

h-4 → 16px
h-8 → 32px
h-16 → 64px
h-32 → 128px
h-64 → 256px

h-1/2 → 50%
h-1/3 → 33.333%
h-2/3 → 66.667%

h-1/4 → 25%
h-2/4 → 50%
h-3/4 → 75%

h-1/5 → 20%
h-2/5 → 40%
h-3/5 → 60%
h-4/5 → 80%

h-1/6 → 16.667%
h-2/6 → 33.333%
h-3/6 → 50%
h-4/6 → 66.667%
h-5/6 → 83.333%

h-full → 100% of its parent



Flexbox with Tailwind

```
<div class="flex justify-center items-center">
```

Tailwind class	CSS equivalent	What it does	
<code>flex</code>	<code>display: flex</code>	Turns the element into a flex container , allowing its children to be arranged using Flexbox.	
<code>flex-row</code>	<code>flex-direction: row</code>	Places children horizontally , from left → right. This is also the default direction.	
<code>flex-col</code>	<code>flex-direction: column</code>	Places children vertically , from top → bottom.	

Typography



```
<h1 class="text-3xl font-bold"> Welcome </h1>  
<p class="text-lg text-gray-600"> Learn Tailwind  
CSS </p>
```

text-sm

text-lg

text-xl

text-3xl

font-normal

font-medium

font-bold

text-center

text-left

text-right

Borders, Rounded Corners & Shadows



Tailwind class	CSS equivalent	Meaning
<code>border</code>	<code>border-width: 1px</code>	Adds a 1px border
<code>border-2</code>	<code>border-width: 2px</code>	Adds a 2px border
<code>border-gray-300</code>	<code>border-color: ...</code>	Sets the border color to gray-300
<code>rounded</code>	<code>border-radius: 0.25rem</code>	Slightly rounds the corners
<code>rounded-lg</code>	<code>border-radius: 0.5rem</code>	Gives the corners a larger rounding
<code>rounded-full</code>	<code>border-radius: 9999px</code>	Makes corners completely rounded ; useful for circles/pills
<code>shadow</code>	<code>box-shadow: ...</code>	Adds a small/default shadow
<code>shadow-md</code>	<code>box-shadow: ...</code>	Adds a medium shadow
<code>shadow-lg</code>	<code>box-shadow: ...</code>	Adds a larger/deeper shadow

Responsive Design



```
<div class="text-sm md:text-lg lg:text-2xl"> Responsive Text </div>
```

Small screen → text-sm Medium screen → text-lg Large screen → text-2xl

sm: md: lg: xl: 2xl:

```
<div class="flex flex-col md:flex-row">
```

 Mobile

[A]

[B]

[C]

 Medium+

[A] [B] [C]

Putting Everything Together



```
<div class="max-w-md mx-auto bg-white p-6 rounded-lg shadow-md">
```

```
<h2 class="text-2xl font-bold text-center mb-4">
```

Student Card

```
</h2>
```

```
<p class="text-gray-600 mb-4">
```

Welcome to the course!

```
</p>
```

```
<button class="w-full bg-blue-500 text-white py-2 rounded-lg">
```

Start Learning

```
</button>
```

```
</div>
```

$\text{max-w-md} = 28\text{rem} = 448\text{px}$

Resources to Learn



Resources to Learn More

- **Websites:**

- [Tailwind CSS Documentation](#)
- [Bootstrap Documentation](#)

